

Does the Assistant Check the Record? Knowledge-Base Poisoning and System-of-Record Validation in Utility Operations RAG

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Abstract—Retrieval-augmented generation (RAG) enables document-grounded technical lookup, but indexed repositories may accept low-trust contributions. We study targeted numeric answer-integrity corruption in utility-operations RAG with GridPoison-100: 35 public-standard items, 45 fictional asset-register items, and 20 legitimate-conflict items. With one inserted document at $k = 5$, the poison was retrieved for 70.0% of public-item queries but 3.3% of asset-item queries. Across Mistral Small 3.2, Qwen3-8B, and GPT-5.6 Luna, exposed public-item trials rarely copied the attacker value at $k = 5$ (CAR 4.1%), yet overall correct-answer rate fell from 75.7–78.6% under clean RAG to 8.6–12.9%. At $k = 2$, where less clean evidence accompanied the poison, public-item CAR rose to 34.6%. In a fixed evidence-rich context with three clean supports and one poison, exact attacker-value adoption was 0%; poison-last instead caused universal non-commitment. A deterministic system-of-record check given the applicable record removed all out-of-range commitments at full coverage, but flagged 64.1–65.7% of otherwise-correct legitimate-conflict answers because most conflict records intentionally held generic profiles. Because violations were rare, defense comparisons are descriptive rather than confirmatory. The results show that exposure, adoption, answer suppression, and record scope must be measured separately.

Index Terms—retrieval-augmented generation, knowledge-base poisoning, power systems, utility operations, large language models, data integrity

I. INTRODUCTION

Large language models are increasingly proposed as natural-language interfaces to technical documentation. In power and utility settings, the most plausible near-term use is not autonomous control but constrained lookup: a field worker asks for an approach distance, a protection engineer checks an interconnection boundary, or a planner retrieves an approved asset limit. RAG makes such answers auditable by grounding them in external documents, yet it also creates a write-path dependency. If an indexed repository accepts contributions from contractors, shared drives, wikis, or externally supplied bulletins, a legitimate user can receive an incorrect value even when the user’s prompt is benign.

Knowledge-base poisoning of RAG is already established [1], [2]. Accordingly, our contribution is not another demonstration that injected text can influence an answer. Instead, we ask what poisoning means when the answer can be graded against an authoritative operational record, whether model prior knowledge protects public facts more than asset-specific facts, and how a separate structured record changes the failure mode.

We study three research questions. **RQ1** asks how targeted poisoning decomposes into retrieval exposure and conditional adoption. **RQ2** asks whether attacker-value adoption differs between public-standard facts and asset-specific facts when exposure is held constant. **RQ3** asks how much system-of-record validation reduces out-of-range commitments relative to two generic defenses, and what false-flag cost it introduces when legitimate documents disagree by jurisdiction or inter-connection.

The contributions are fourfold: (1) a write-path threat model for low-trust utility-document repositories; (2) GridPoison-100, a released benchmark with source-grounded public items, a fictional asset register, and legitimate-conflict controls; (3) a two-stage evaluation that separates retrieval compromise, conditional adoption, and consequence grading; and (4) an accounting of what deterministic system-of-record validation gains and costs under partial coverage and deliberately simplified record scope.

II. RELATED WORK

PoisonedRAG showed that a small number of attacker-controlled texts can steer retrieval-augmented models toward targeted answers [1]. Subsequent work has examined retrieval, database, and generator factors that influence poisoning exposure and attack success [2]. Because those effects are already known, retrieval depth is a secondary factor here rather than a novelty claim.

RobustRAG introduced an isolate-then-aggregate strategy for retrieval corruption [3]; we use this family as a generic defense baseline rather than presenting it as new. ClashEval quantified conflicts between parametric knowledge and retrieved evidence [4]. Other work has shown that evidence phrasing, publication time, and metadata can affect model preferences [5], [6]. These findings motivate our vendor-bulletin condition but also rule out treating authority framing as a primary contribution.

LLM risks in power systems have already been discussed and experimentally explored [7], [8]. Our study therefore avoids claims of being the first LLM-security work in the domain.

III. THREAT MODEL

The attacker is an authorized but low-trust writer, such as a contractor with write access to an indexed collaboration repository, or the sender of a spoofed technical bulletin

that is subsequently indexed. The attacker may add one to three new documents but cannot edit or delete existing documents. They cannot modify the structured asset/settings record, EMS/SCADA, protection devices, model weights, or the user’s query. The attacker knows the target asset or topic and can read existing documents on it, but does not know the exact query wording, embedding model, or retrieval results.

The attack goal is targeted answer-integrity corruption: a legitimate lookup should commit to an attacker-chosen numeric value. The primary poison is a factual statement rather than an instruction to the model, separating this setting from indirect prompt injection. Confidentiality, model-weight poisoning, multi-turn social engineering, and physical control-loop attacks are outside scope. Denial of service is not an attack goal, but we report loss of correct answers because it is the dominant observed effect.

IV. GRIDPOISON-100

GridPoison-100 contains 100 lookup items divided into three tiers. Tier 1 contains 35 public normative facts drawn from Massachusetts DER settings and return-to-service material, OSHA 29 CFR 1910.269 minimum-approach-distance tables, and NERC PRC-024-3 interconnection boundaries [9], [10], [11], [12]. Each public item records a source URL, source locator, units, applicability conditions, and an allowable interval.

Tier 2 contains 45 asset-specific facts from a declared fictional utility register. Asset identifiers and values are synthetic by construction, preventing a model from knowing them from pretraining. The register covers transformer emergency ratings, breaker current ratings, feeder thermal limits, BESS continuous discharge limits, and cable ampacity values. For Tier 2, the register is the authoritative record by construction; no claim is made that these values describe a real network.

Tier 3 contains 20 legitimate-conflict items. Two documents can disagree because of jurisdiction, interconnection, category, or table scope, while the query explicitly names which context applies. These items are not poisoned; they measure whether a defense incorrectly flags a correct context-specific answer.

The clean corpus contains 400 short template-generated documents: three support documents per poisonable item in three styles (reference note, lookup summary, and training card), 40 legitimate-conflict documents, and 120 on-topic distractors. Documents carry item and asset identifiers, so the corpus is cleaner and more explicit than a real utility repository. Each poison is a minimal pair of one support document that changes only the critical value, making the conflict maximally visible. The main poison moves the value outside the encoded benchmark interval; a wrong-but-in-range variant separates factual incorrectness from interval violation, and a dated vendor-bulletin form tests authority framing.

V. EXPERIMENTAL DESIGN

A. Reference conditions and models

B0 is closed-book generation, B1 is clean RAG, and B2 is poisoned RAG with the same base prompt. The base prompt

requests one committed numeric value when evidence is unambiguous and `VALUE: UNKNOWN` when supplied sources are inconsistent; B2 therefore already permits abstention on conflict, while D1 adds an explicit warning that retrieved sources may be falsified. Retrieval uses sentence-transformers/all-MiniLM-L6-v2 embeddings with cosine similarity and one document per chunk. The evaluated models are Qwen3-8B, Mistral Small 3.2 24B, and GPT-5.6 Luna, called through OpenRouter at temperature zero with a 256-token output cap. Provider routing was not pinned; resolved model/provider metadata, response ID, token usage, and timestamp were logged per response.

B. Stage A: retrieval exposure

RQ1 uses natural dense retrieval. The lower-strength condition inserts one out-of-range poison; the stress condition inserts three distinct poison paraphrases. `Poison@k` is reported separately from generation behavior. A retrieval-only pre-generation calibration found that one-document exposure was strongly imbalanced between Tier 1 and Tier 2, which would confound a direct cross-tier adoption comparison.

C. Stage B: forced exposure

RQ2 therefore uses a fixed five-passages context rather than natural retrieval. Each item receives three clean same-item support passages, one deterministic on-topic distractor, and exactly one poison. The poison is counterbalanced in the first or last position. This deliberately evidence-rich design holds exposure and context length constant while comparing public-standard and asset-specific facts; it is not intended to represent natural attack strength.

D. Defenses

D1 is a cautious prompt instructing the model to identify unresolved source conflicts. D2 isolates each retrieved passage, obtains one value per passage with the cautious prompt, and returns the plurality value; ties yield `UNKNOWN`. It is a simplified instance of the isolate-then-aggregate family [3], not RobustRAG’s certified aggregation. D3 is deterministic system-of-record validation: if the item’s record is covered and the committed value differs from it, the answer is replaced by a verification flag. D3 is given the applicable record key, so query-to-record linking errors are not modeled. For 78 of the 80 poisonable items the record equals the benchmark target; the exceptions are two Massachusetts undervoltage settings for which the simplified record stores IEEE 1547-2018 Category II defaults [13], [11]. For 13 of the 20 Tier 3 items the record stores a generic alternative rather than the applicable contextual variant. D3 therefore approximates an answer-key check on poisonable items; its purpose here is to quantify an upper-bound filtering benefit and the cost of record-scope mismatch, not to propose a new defense algorithm. Coverage at 40%, 70%, and 100% is fixed deterministically by hashing item identifiers.

RQ3 uses the three-poison $k = 5$ stress condition to avoid a retrieval floor when comparing defenses. Secondary one-factor

deviations examine a dated vendor bulletin, a wrong-but-in-range value, and $k = 2$. No full factorial sweep is performed.

VI. METRICS AND STATISTICAL ANALYSIS

The raw logs retain labels for correct (**C**), attacker-value adoption (**A**), another committed value (**W**), conflict flag (**F**), and refusal (**R**). For reported metrics, numeric C/A/W grading is deterministic from the required `VALUE`: line and item tolerance, while every response without a parsed numeric value is pooled as non-commitment (**N**); we do not interpret the parser-level F/R split.

Let E_t indicate whether a poisoned document occurs in the retrieved top- k context. We report Poison@ k , attack success rate (ASR), conditional adoption rate (CAR), correct-response rate (CRR), non-commitment rate (NCR), and benchmark-violation rate (BVR):

$$\text{CAR} = P(A \mid E_t = 1), \quad (1)$$

$$\text{BVR} = \frac{1}{|T|} \sum_t \mathbb{1}[\hat{v}_t \notin [L_i, U_i]]. \quad (2)$$

BVR counts only committed numeric values; non-committal outputs are non-violations by definition, so BVR is reported together with CRR. It is a benchmark consequence metric, not a physical grid-safety measure.

The item is the unit of analysis and the two query phrasings are repeated observations. Rates receive 95% item-cluster bootstrap intervals with 2,000 resamples; zero-event Stage-B adoption cells additionally receive exact Clopper–Pearson bounds. The locked primary RQ3 comparison is D3 at 70% coverage versus D1 and D2. Paired comparisons use item-level exact McNemar tests, with an item counted as violating only when both phrasings violate, and Holm correction over the six primary comparisons; comparisons with vanilla RAG are secondary and descriptive. Tier 3 BFFR is conditioned on clean-RAG correctness so D3 is charged only when it flags an otherwise-correct context-specific answer.

VII. RESULTS

Clean-RAG CRR was 87.8% on Tier 2 (pre-specified threshold: 80%) and 75.7–78.6% on Tier 1.

A. Natural-retrieval behavior (RQ1)

Table I reports retrieval exposure and generation outcomes. Exposure is deterministic for the shared retriever; CRR and CAR are ranges across models. At $k = 5$, one poison was retrieved in 49 Tier 1 trials per model; only 0–3 remained correct, while direct attacker adoption was 4.1% of exposed trials. Reducing retrieval depth to $k = 2$ changed the evidence balance: all three models adopted the attacker value in 9 of 26 exposed Tier 1 trials (CAR 34.6%). The wrong-but-in-range poison was copied in only 1–2 of 35 exposed trials per model.

B. Fixed-context exposure (RQ2)

In the evidence-rich fixed context (three clean same-item supports, one distractor, one poison), attacker-value adoption and BVR were 0% in every model/tier/position cell. With zero events, the exact 95% upper bounds on per-item adoption are 10.0% for Tier 1 ($n = 35$) and 7.9% for Tier 2 ($n = 45$). Table II reports correct-answer retention. Every poison-last trial produced no numeric commitment. Because the base prompt explicitly permits `VALUE: UNKNOWN` when evidence is inconsistent, this is a conflict-handling result rather than evidence of inherent poisoning robustness.

C. Record-check accounting (RQ3)

The locked primary comparison is D3 at 70% coverage versus D1 and D2. Table III reports paired BVR reduction in percentage points; positive values favor D3. Against vanilla poisoned RAG, D3@70 removed 6/7, 5/6, and 6/8 violating trials for Mistral, Qwen, and GPT-5.6 Luna, respectively, and full coverage removed all of them. D2 increased Qwen BVR by 3.1 points relative to B2. Only 5–8 trials violated per model and the primary McNemar comparisons had at most 3 discordant items; all six Holm-adjusted tests therefore remained non-significant ($p_{\text{Holm}} = 1.0$). The defense comparisons are descriptive.

D. Legitimate-conflict cost

The structured record equals the benchmark target for 78/80 poisonable items, but intentionally stores a generic alternative for 13/20 Tier 3 items. BFFR therefore measures a controlled record-scope gap and is conditioned on clean-RAG correctness. At 70% coverage, conditional BFFR was 43.6–45.7%; at full coverage it was 64.1–65.7%. The cost is therefore driven by record scope, not only by model behavior.

VIII. DISCUSSION

Natural retrieval reveals a stronger failure mode than the forced-exposure headline alone. At $k = 5$, a retrieved poison rarely transferred its exact value, but exposed trials usually lost the correct answer and became non-committal. When retrieval depth was reduced to $k = 2$, Tier 1 CAR rose to 34.6%, showing that direct adoption depends strongly on how much competing clean evidence accompanies the poison. For this corpus, poisoning therefore manifests primarily as answer suppression at $k = 5$ and as direct corruption when evidence is sparser.

The fixed-context RQ2 control produced zero exact adoption, and poison-last caused universal non-commitment. That result should not be read as inherent model robustness: the context contains three clean same-item supports against one poison, and the base prompt explicitly instructs abstention on unresolved conflict. Moreover, closed-book Tier 1 CRR was only 0–2.9%, so the planned hypothesis that parametric knowledge would preferentially protect public standards is not supported; the tested models showed little exact prior knowledge of those values.

TABLE I
NATURAL-RETRIEVAL OUTCOMES (%; $k = 5$ UNLESS NOTED).

Condition	Exp.	Tier 1 (public)		Exp.	Tier 2 (asset)	
		CRR	CAR		CRR	CAR
Clean RAG	–	75.7–78.6%	–	–	87.8%	–
1 poison	70.0%	8.6–12.9%	4.1%	3.3%	84.4%	0.0%
3 poisons	70.0%	7.1–15.7%	4.1%	76.7%	14.4%	4.3–5.8%
1 bulletin	67.1%	20.0–22.9%	6.4–8.5%	7.8%	80.0%	0.0–14.3%
1 poison, $k = 2$	37.1%	14.3%	34.6%	3.3%	62.2%	0.0%

TABLE II
CORRECT-ANSWER RATE UNDER FIXED EXPOSURE (%).

Model	First/T1	First/T2	Last/T1	Last/T2
Mistral Small 3.2	8.6%	97.8%	0.0%	0.0%
Qwen3-8B	42.9%	57.8%	0.0%	0.0%
GPT-5.6 Luna	74.3%	28.9%	0.0%	0.0%

D3 measures a different mechanism. Because it is given the applicable record key and the record equals the target for 78/80 poisonable items, its benefit is largely an accounting of how many existing violations fall on covered items rather than a change in language-model behavior. At 70% coverage it removed most B2 violations and full coverage removed all of them, as expected, but the locked D3-vs-D1/D2 comparisons involve only a handful of violating items and have little inferential power. The result is therefore best interpreted as an upper-bound record-check benefit under ideal linking.

The cost is equally structural. Thirteen of 20 legitimate-conflict records intentionally store a generic profile rather than the applicable contextual variant. As coverage grows, D3 increasingly flags correct jurisdiction- or interconnection-specific answers. The operational lesson is consequently about record scope: a structured check can be useful only when its authority, context, and escalation policy are explicit.

IX. LIMITATIONS

GridPoison-100 is a controlled benchmark rather than a deployed utility corpus. The documents are template-generated, carry explicit identifiers, and the poisons are minimal pairs that make conflicts maximally visible; together with the base prompt’s abstention instruction, these choices favor detection over attacker-value adoption. The study uses one dense retriever, three model families, an unpinned OpenRouter serving route, and scalar lookup tasks rather than procedural switching or open-ended engineering judgment. GPT-5.6 Luna produced 50 empty responses under the 256-token cap, including four in Stage B; these are counted as non-correct and non-committal.

Tier 1 targets and Tier 3 alternatives were checked against the cited primary sources, while Tier 2 values are fictional by declaration and authoritative only within the synthetic register. The encoded intervals are benchmark definitions derived from those references or records, not engineering operating limits. No power-flow, protection, thermal, or transient simulation is performed, so BVR does not quantify physical consequences.

D3 is given the applicable record key and approximates an answer-key check on poisonable items, so its measured benefit is an upper bound that excludes query-to-record linking errors. Finally, the fixed Stage-B context is deliberately evidence-rich, so its zero-adoption result should not be generalized to weaker or noisier evidence settings.

X. CONCLUSION

GridPoison-100 shows that retrieval exposure, attacker-value adoption, answer suppression, and benchmark consequence are distinct outcomes in utility-operations RAG. In this study a retrieved poison at $k = 5$ rarely transferred its value but usually displaced the correct answer, while direct adoption rose sharply when less clean evidence accompanied the poison at $k = 2$. A structured record check removed out-of-range commitments on covered items under ideal record linking, but also flagged correct answers whenever the simplified record lacked the applicable contextual variant. The practical implication is not that more filtering is always better, but that retrieval integrity and record scope must be evaluated together under the stated threat model.

GENERATIVE AI USE DISCLOSURE

Generative AI systems including ChatGPT (OpenAI), Claude (Anthropic), and Grok (xAI) assisted software scaffolding, benchmark/corpus drafting, literature triage, source-consistency checking, adversarial methodological review, and language editing. All reported numerical results derive from logged experiments and deterministic analysis code; the authors remain fully responsible for the accuracy, originality, and integrity of the manuscript.

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TABLE III
PRIMARY RQ3 COMPARISON AND VIOLATION COUNTS. BRACKETS ARE 95% ITEM-CLUSTER BOOTSTRAP INTERVALS.

Model	D3@70 vs D1 (pp)	D3@70 vs D2 (pp)	B2 → D3@70	D3@100
Mistral Small 3.2	2.5 [-0.6, 6.2]	3.8 [0.6, 8.1]	7 → 1	0
Qwen3-8B	3.1 [0.6, 6.9]	6.2 [1.9, 11.2]	6 → 1	0
GPT-5.6 Luna	3.8 [0.6, 7.5]	3.8 [0.0, 8.8]	8 → 2	0

TABLE IV
CONDITIONAL BENIGN-CONFLICT FALSE-FLAG RATE (BFFR).

Model	40%	70%	100%
Mistral Small 3.2	17.9%	43.6%	64.1%
Qwen3-8B	17.1%	45.7%	65.7%
GPT-5.6 Luna	17.9%	43.6%	64.1%

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